

Birla Institute of Technology and Science, Pilani.

Mid Semester Examination: CS/ECE/EEE/INSTR F215: Digital Design

Marks: 90

AY: 2024-25, Semester: I

Date: 7-October-2024, Monday

Time: 90 minutes

CLOSED BOOK

Pages:04

BITS ID:_____Name:_____

Q1 Design a non-sequential counter ($Q_A Q_B Q_C$; Q_A : **MSB**; Q_C : **LSB**) that counts 4-2-1-0-5-3 and [16]

repeat. Design the counter assuming the next state for unused states, as don't care.

Components available for design are ONE T-flip-flop, TWO JK flip-flop, and ONE 4:1 MUX.

(Note: Only the final circuit needs to be shown)

Q2 Realize the function $F(A, B, C, D) = \sum m(0, 2, 3, 5, 6, 8, 9) + \sum d(10, 11, 12, 13, 14, 15)$. Use AB as a select input to MUX (s1 s0), respectively. Only TRUE inputs are available for design. Choose the required components from the following list. [20]

- **ONE** 2-to-4-line decoder with active high output and one active-high enable.
- **ONE** 4-to-2-line encoder with active high enable.
- **ONE** 4:1 MUX with active high enable.
- **ONE** 1:4 DMUX with active high enable.

- Q3 A system to detect prime numbers has a 1-digit BCD (PQRS; **P: MSB**, S: LSB) as input and 1-bit Z as output. Z will be 1 (high) if the input is a prime number and 0 (low) if the input is not a prime number. **Note:** 0 and 1 are not prime numbers. Plot the K-map for Z. List all the Prime Implicants (PIs), Essential Prime Implicants (EPIs), and all minimum SOP forms. Realize the obtained expression using minimum numbers of 2-input NAND gates only. TRUE and COMPLEMENT inputs are available for design. [20]

List of PI's	List of EPI's

Minimum SOP form: _____

Realization:

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- Q4 Realize the function $F(w, x, y, z) = \sum m(2, 3, 4, 5, 6, 8, 9) + \sum d(10, 11, 12, 13, 14, 15)$ [14]

The components available for design are **ONE** 1-bit full adder, **ONE** 1-bit full subtractor, and **ONE** 1-bit half subtractor. Only True inputs are available.

Q5 Design a sequence detector (overlapping) circuit that detects the sequence 1001. The circuit [20]
should have a 1-bit serial input, X, and two 1-bit outputs, Z1 and Z2. **Z1** becomes high for one
clock cycle whenever the sequence 1001 is detected. **Z2** becomes high when the sequence
1001 is detected **TEN** times. After detecting the sequence ten times, the operation stops.
Choose the required components from the following list.

- **ONE** 4-bit binary UP counter with active-high enable with clear and preset.
- **ONE** 4-bit shift register with serial-in, parallel-out, and active low-enable.
- **ONE** 4-to-16-line decoder with active high output and one active-high enable
- **ONE** 1:16 DMUX with active high enable